STAGE 03

Project Design Phase

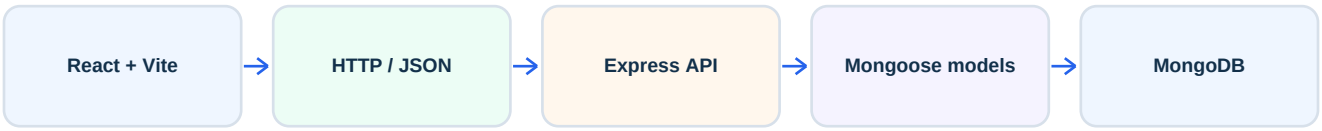
Architecture, conceptual data model, API flow, and UI patterns.

PROJECT	STACK	DOCUMENT
HotelPro	React • Vite • Express • MongoDB	Stage 3 of 7

Documentation note

Prepared from HotelPro project context shared earlier. Verify implementation details against the current repository. Historical test/build figures are marked as reported, not freshly verified.

Architecture



Layer responsibilities

Layer	Responsibilities	Design checks
Client	Pages, forms, tables, navigation	No UI-only security
Express	Routes, validation, auth, business rules	Consistent errors/status
Mongoose	Schemas and persistence	Validation/references
MongoDB	Application records	Access control and backups

Conceptual data model

Entity	Example fields	Rule / relationship
User	Name, email, password hash, role	Can be associated with bookings
Room	Number/type, price, capacity, status	Referenced by bookings
Booking	User, room, dates, status, total	Validate date range/availability
Payment	Booking, amount, status, reference	Avoid raw card details

Request lifecycle

UI design checks

Screen	Recommendation
Dashboard	Clear KPI cards and recent activity; no fake metrics as real
Room table	Status, price, capacity, loading/empty/error states
Booking form	Date validation and availability checks
Admin/staff	Role-specific actions, protected server-side
Responsive UI	Mobile width, focus, labels